

By [Lareina Yee](#), [Michael Chui](#), [Roger Roberts](#), and [Sven Smit](#)

Which frontier technologies matter most for companies in 2025? Our annual tech trends report highlights the latest technology breakthroughs, talent trends, use cases, and their potential impact on companies across sectors.

DOWNLOADS



McKinsey Technology Trends Outlook 2025

[↓ Full Report \(108 pages\)](#)

The global technology landscape is undergoing significant shifts, propelled by fast-moving innovations in technologies. These are exponentially increasing demand for computing power, capturing the attention of management teams and the public, and accelerating experimentation. These developments are occurring against a backdrop of rising global competition as countries and corporations race to secure leadership in producing and applying these strategic technologies.

This year's McKinsey Technology Trends Outlook provides in-depth perspectives on 13—a “baker's dozen”—frontier technology trends with the potential to transform global business. Executives today face a mandate to navigate rising complexity, scale emerging s ^{BETA} [Ask](#) ^{IC} [McKinsey](#) build trust in a world where the lines between digital and physical and centralize decentralized continue to blur. The insights in this report can help business leaders decide

which of these frontier technologies are most relevant to their companies by demonstrating how others are starting to apply them today. These findings emerge from our analysis of quantitative measures of interest, innovation, equity investment, and talent that underpin each of the 13 trends and explore the underlying technologies, uncertainties, and questions around them. (For more about our research, please see the sidebar, “Research methodology.”)

This outlook highlights transformative trends that are driving innovation and addressing critical challenges across sectors. Artificial intelligence stands out not only as a powerful technology wave on its own but also as a foundational amplifier of the other trends. Its impact increasingly occurs via a combination with other trends, as AI both accelerates progress within individual domains and unlocks new possibilities at the intersections—accelerating the training of robots, advancing scientific discoveries in bioengineering, optimizing energy systems, and much more. The evolution of AI solutions in the marketplace increasingly combines aspects of trends we previously analyzed separately as applied AI and generative AI, so this year, they are examined together.

Even as excitement about AI applications and their use cases builds, realizing AI’s full potential across sectors will require continued innovations to manage computing intensity, reduce deployment costs, and drive infrastructure investment. This will also demand thoughtful approaches to safety, governance, and workforce adaptation, creating a wide range of opportunities for industry leaders, policymakers, and entrepreneurs alike.

New and notable

In addition to the growing reach of AI, another new trend we have chosen to highlight in this year’s report is [agentic AI](#), which has rapidly emerged as a major focus of interest and experimentation in enterprise and consumer technology. Agentic AI combines the flexibility and generality of AI foundation models with the ability to act in the world by creating “virtual coworkers” that can autonomously plan and execute multistep workflows. Although quantitative measures of interest and equity investment levels are as yet relatively low compared with more established trends, agentic AI is among the fastest growing of this year’s trends, signaling its potentially revolutionary possibilities.

AI is also the primary catalyst for another trend we highlight this year: application-specific semiconductors. While [Moore’s Law](#) and the semiconductor layer of the technology stack have long been key enablers of other tech trends, innovations in semiconductors have spiked as reflected in quantitative metrics such as number of patents. These innovations have come in response to exponentially higher demands for computing capacity, memory, and networking for AI training and inference, as well as a need to manage cost, heat, and electric power consumption. This has given rise to a slew of new products, new competitors, and new ecosystems.

Technology trends also have a variety of profiles along the dimensions we analyzed. AI is a widely applicable, general-purpose technology with use cases in every industry and business function—and thus lots of innovation and interest—and it is scaling rapidly across the business landscape. Quantum technologies have a different profile. Quantum computing has the potential for transformative impact in certain critical domains, such as cryptography and material science, and the basic technology continues to be developed. Recent announcements, particularly by technology giants, have sparked increased interest, but real-world business impact will require even more technology advancements to make quantum computing practical. Other trends and subrends vary across the multiple dimensions we analyzed, offering different approaches—from watchful waiting to aggressive deployment—to business leaders depending on their industries and competitive positions.

From the rise of robotics and autonomous systems to the imperative for responsible AI innovations, this year's technology developments underscore a future where technology is more adaptive, collaborative, and integral to solving global problems. This is illuminated by themes that cut across trends this year:

- *The rise of autonomous systems.* Autonomous systems, including physical robots and digital agents, are moving from pilot projects to practical applications. These systems aren't just executing tasks; they're starting to learn, adapt, and collaborate. Autonomy is moving toward broad deployment, whether through coordinating last-mile logistics, navigating dynamic environments, or acting as virtual coworkers, among other skills.
- *New human-machine collaboration models.* Human-machine interaction is entering a new phase defined by more natural interfaces, multimodal inputs, and adaptive intelligence. From immersive training environments and haptic robotics to voice-driven copilots and sensor-enabled wearables, technology is becoming more responsive to human intent and behavior. This evolution is shifting the narrative from human replacement to augmentation—enabling more natural, productive collaboration between people and intelligent systems. As machines get better at interpreting context, the boundary between operator and cocreator continues to dissolve.
- *Scaling challenges.* The surging demand for compute-intensive workloads, especially from gen AI, robotics, and immersive environments, is creating new demands on global infrastructure. Data center power constraints, physical network vulnerabilities, and rising compute demands have exposed cracks in global infrastructure. But the challenge isn't just technical: Supply chain delays, labor shortages, and regulatory friction around grid access and permitting are slowing deployments. As a result, scaling now means solving not only for technical architecture and efficient design but also for the messy, real-world challenges in talent, policy, and execution.
- *Regional and national competition.* Global competition over critical technology intensified. Countries and corporations have doubled down on sovereign infrastructure,

localized chip fabrication, and funding technology initiatives such as quantum labs. This push for self-sufficiency isn't just about security; it's about reducing exposure to geopolitical risk and owning the next wave of value creation. The result is a new era of tech-driven competition where nations have a stake in critical industries.

- *Scale and specialization are growing simultaneously.* Growth on these vectors is enabled by innovation in cloud services and advanced connectivity. On one hand, we see rapid growth in general-purpose model training infrastructure in vast, power-hungry data centers, while on the other, we observe accelerating innovation “at the edge,” with lower-power technology embedded in phones, cars, home controls, and industrial devices. This is creating ecosystems that deliver massive large language models with staggering parameter counts, as well as a growing range of domain-specific AI tools that can run almost anywhere. Leaders will balance centralized scale with localized control: Think modular microgrids for clean energy or bespoke robotics for niche manufacturing.
- *Responsible innovation imperatives.* As technologies become more powerful and more personal, trust is increasingly the gatekeeper to adoption. Companies face growing pressure to demonstrate transparency, fairness, and accountability, whether in AI models, gene editing pipelines, or immersive platforms. Ethics are no longer just the right thing to do but rather strategic levers in deployment that can accelerate—or stall—scaling, investment, and long-term impact.

The following illustrations show how different frontier technologies can work together to provide innovative solutions in the future:

Factory machine repair



Personalized medicine delivery



Wind farm



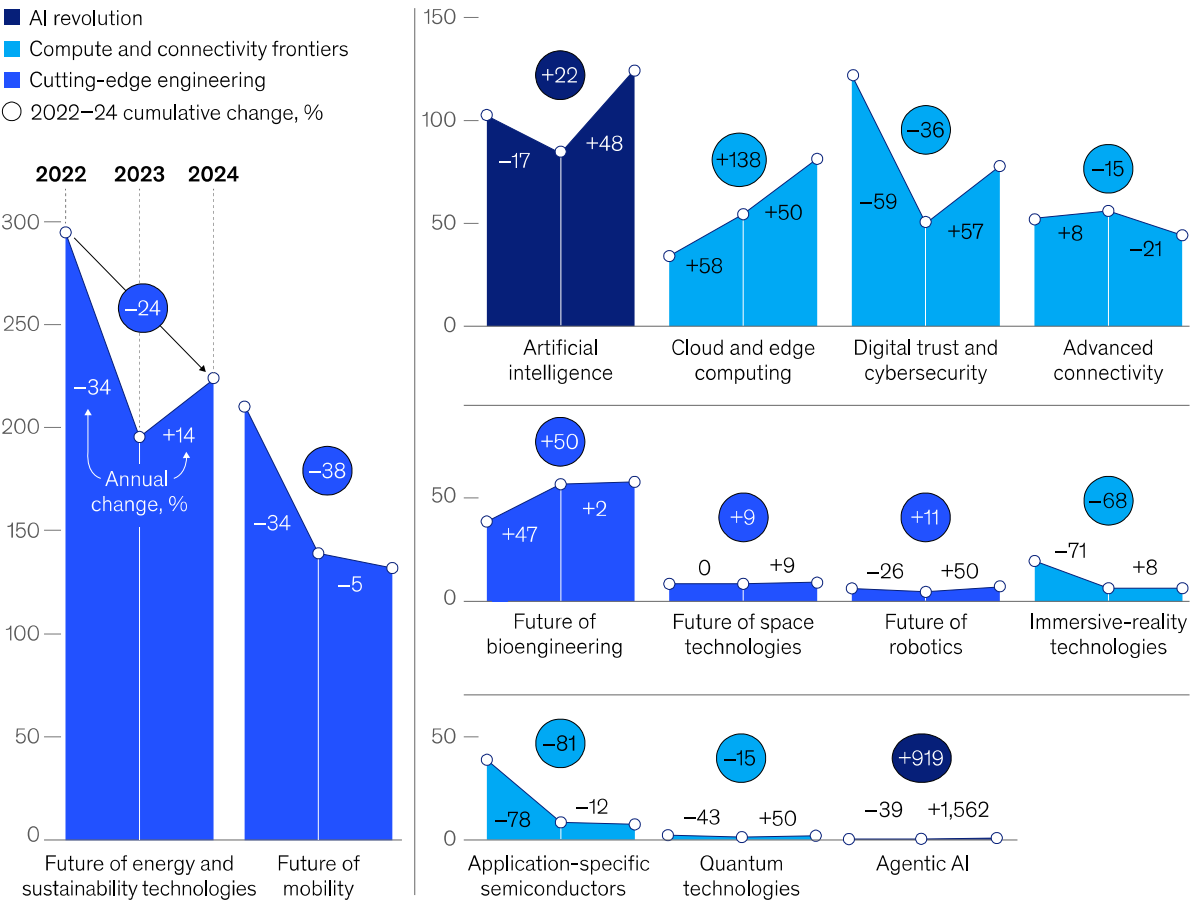
Illustrations: T.M. Detwiler

Levels of equity investment in trends such as cloud and edge computing, bioengineering, and space technologies increased despite the broader market dip in 2023, while investments in other trends, such as AI and robotics, dipped only to recover to higher levels in 2024 than they achieved two years prior. The two trends with the highest levels of equity investment, the future of energy and sustainability technologies and the future of mobility, declined overall in 2023, but the former bounced back in 2024 (exhibit).

Exhibit

Equity investments increased in ten of 13 technology trends in 2024.

Trend investments, 2022–24, \$ billion



Note: Data includes private-market and public-market capital raises across venture capital and corporate and strategic M&A (including joint ventures), private equity investments (including buyouts and private investment in public equity), and public investments (including IPOs). Excludes corporate capital and operational expenditures.
Source: PitchBook; McKinsey analysis

McKinsey & Company

Our baker's dozen of technology trends shaping 2025 underscores the vast potential of emerging technologies and the need for strategic alignment in an AI-powered future. For executives, success will hinge on identifying high-impact domains in which they trends, investing in the necessary talent and infrastructure, and addressing exte



regulatory shifts and ecosystem readiness. By fostering collaboration, bridging ecosystem gaps, and maintaining a long-term vision, leaders can accelerate adoption and position their organizations to drive the next wave of technological transformation. Those who act with focus and agility will not only unlock new value but also shape the future of their industries and the future of today's emerging frontier technologies.

How relevant and useful is this article for you?

☆☆☆☆

ABOUT THE AUTHOR(S)

Lareina Yee is a McKinsey Global Institute director and a senior partner in McKinsey's Bay Area office, where **Michael Chui** is a senior fellow at QuantumBlack, AI by McKinsey, and **Roger Roberts** is a partner at QuantumBlack, AI by McKinsey; **Sven Smit** is chair of the McKinsey Global Institute and a senior partner in the Amsterdam office.

The authors wish to thank the following McKinsey colleagues and alumni for their contributions to this research: Aamer Baig, Ahsan Saeed, Alex Singla, Alexander Sukharevsky, Alex Zhang, Alizee Acket-Goemaere, Amishi Bharti, Amy Silverstein, Andrea Del Miglio, Andreas Breiter, Andreas Schlosser, Ani Kelkar, Anna Heid, Anu Madgavkar, Arjita Bhan, Bernd Heid, Bharath Aiyer, Bill Gregg, Bill Wiseman, Brooke Stokes, Bryan Richardson, Charlie Lewis, Christian Staudt, Clint Wood, Daniel Herde, Daniel Wallance, David Naney, Delphine Nain Zurkiya, Diana Tang, Egor Kiselev, Eliza Spinna, Emily Shao, Erika Stanzl, Fabian Queder, Gabriel Morgan Asaftei, Giacomo Gatto, Godart van Gendt, Hamza Khan, Henning Soller, Ichiro Otobe, Jacob Achenbach, Jakob Fleischmann, Jawad Mourabet, Jeffrey Caso, Jenny Tran, Jesse Noffsinger, Jim Adams, Jim Boehm, Jonathan Tilley, Joshua Katz, Justin Greis, Karl Grosselin, Kersten Heineke, Kevin F. Lu, Kitty Lakner, Klaus Pototzky, Klemens Hjartar, Luca Bennici, Marc Sorel, Mark Patel, Markus Wilthaner, Martin Harrysson, Martin Kellner, Martin Wrulich, Matt Higginson, Medha Bankhwal, Mekala Krishnan, Michael Bogobowicz, Nandika Komirisetti, Naveen Sastry, Olivia White, Paolo Spranzi, Prasad Ganorkar, Ryan Brukardt, Sebastian Mayer, Sian Griffiths, Sonja Lindberg, Soumya Banerjee, Stefan Burghardt, Stephen Xu, Tapio Melgin, Tarik Alatovic, Thomas Hundertmark, Tom Brennan, Wendy Zhu, Yaman Tandon, Yvonne Ferrier, and Zina Cole.

Special thanks to McKinsey Global Publishing colleagues Daniel Eisenberg, Diane Rice, Janet Michaud, Juan M. Velasco, Kanika Punwani, LaShon Malone, Mary Gayen, Michael Goesele, Nayomi Chibana, Rachel Robinson, Regina Small, Stephanie Strom, Stephen Landau, and Victor L. Cuevas for making this report come alive.

EXPLORE A CAREER WITH US

[Search openings](#)

BETA
Ask ☆
McKinsey